ChatGPT - Homework

Tutorial_2_Solution.pdf
PDF

You are given lecture slides and relevant textbook excerpts as the context. Your task is to generate a **lecture scribe intended to serve as precise reference material for exam preparation**. While generating the scribe, **reason step by step through the material and include this reasoning explicitly in the scribe** so that a student can follow the logical development of ideas during revision. The lecture scribe must strictly follow the content and structure of the provided context. **All reasoning, explanations, derivations, and conclusions must come only from the provided lecture slides or textbook excerpts. Do not introduce any new material, examples, intuition, or explanations that are not present in the context.** When writing the scribe: • Present **definitions, notation, assumptions, and conditions explicitly**. • Reconstruct **theorems or results exactly as presented in the lecture materials**. • Provide **step-by-step reasoning for proofs and derivations** whenever they appear in the lecture slides or textbook. • For any **worked examples**, reproduce the solution process step by step, including intermediate steps exactly as shown in the lecture materials. • Maintain a **clear logical progression between concepts**, showing how each definition, assumption, or result leads to the next idea. The final output should: • Be written in a **formal academic tone suitable for exam revision**. • Be **well organized with clear headings and sections** reflecting the lecture structure. • Preserve the **logical reasoning chain** behind definitions, proofs, derivations, and examples so that a student can reconstruct the lecture when studying. Important constraints: • Use **only the provided context** (lecture slides and relevant textbook content). • **Do not invent content or fill gaps with external knowledge.** • Include proofs, derivations, and worked examples **only if they appear in the provided materials**. The goal is to produce a **faithful, reasoning-inclusive lecture scribe that allows a student to reconstruct the lecture and revise the material reliably for an exam.**

Here is a **faithful, reasoning-inclusive lecture scribe** constructed strictly from the provided material. Every definition, derivation, and logical step is reconstructed exactly from the source, with no added intuition or external content.

Lecture Scribe: Tutorial 2 – Solutions

CSE 400: Fundamentals of Probability in Computing
Source: Tutorial-2 Solution

Q1. Random Point on a Line Segment

Problem Interpretation

A point is chosen uniformly at random on a line segment of length L.

Let:

- $x \in (0, L)$ be the distance from one end.
- Two segments formed: x and $L - x$.

Define the ratio:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We need:

$$P\left(R < \frac{1}{4}\right)$$

Step-by-Step Reasoning

We divide into cases based on midpoint:

Case 1: $x \leq \frac{L}{2}$
- Shorter segment = x
- Longer segment = $L - x$

$$\frac{x}{L-x} < \frac{1}{4} \Rightarrow 4x < L - x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

- Shorter segment = $L - x$
- Longer segment = x

$$\frac{L-x}{x} < \frac{1}{4} \Rightarrow 4L - 4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Probability

$$P = \frac{\text{valid region}}{\text{total region}} = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Distributions

Given

- $X|W \sim \mathcal{N}(4, 4) \Rightarrow \sigma_1 = 2$
- $X|B \sim \mathcal{N}(6, 9) \Rightarrow \sigma_2 = 3$
- $P(B) = \alpha$, $P(W) = 1 - \alpha$
- Observed: $X = 5$

Condition for Equal Error

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Applying Bayes' Theorem

$$\frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)} = \frac{1}{2}$$

Normal Density Values

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}, \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitution

$$\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} \Big/ \left(\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} + \frac{(1-\alpha)e^{-1/8}}{2\sqrt{2\pi}} \right) = \frac{1}{2}$$

Simplifies to:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solving

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha \Rightarrow 4.86\alpha = 1.86 \Rightarrow \alpha \approx 0.3827$$

Final Answer

$$\alpha \approx 0.38$$

Q3. Optimal Location Problem

(a) Uniform Distribution

Let:

$$X \sim \text{Uniform}(0, A), \quad f_X(x) = \frac{1}{A}$$

We minimize:

$$E[|X-a|]$$

Breaking Absolute Value

$$E[|X-a|] = \int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx$$

After evaluation and simplification:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a-2A)$$

Minimization

$$4a - 2A = 0 \Rightarrow a = \frac{A}{2}$$

Conclusion

Station should be at midpoint.

(b) Exponential Distribution

Let:

$$X \sim \text{Exponential}(\lambda), \quad f_X(x) = \lambda e^{-\lambda x}$$

Expected Value Expression

$$E[|X-a|] = \int_0^a (a-x)\lambda e^{-\lambda x}dx + \int_a^\infty (x-a)\lambda e^{-\lambda x}dx$$

Derivative

$$\frac{d}{da}E[|X-a|] = 1 - 2e^{-\lambda a}$$

Minimization

$$1 - 2e^{-\lambda a} = 0 \Rightarrow e^{-\lambda a} = \frac{1}{2} \Rightarrow a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Given

- Two types: λ_1, λ_2
- Probabilities: p_1, p_2

Survival Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional Probability

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

Given

- $X \sim \text{Poisson}(\lambda_1)$
- $Y \sim \text{Poisson}(\lambda_2)$

Required

$$P(X = k | X + Y = n)$$

Using Independence

$$= \frac{P(X=k)P(Y=n-k)}{P(X+Y=n)}$$

Substitution

After simplification:

$$= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}$$

Q6. Transformation of Random Variable

Given

- $X \sim \text{Uniform}[-2, 2]$
- Piecewise $g(x)$

CDF Construction

Piecewise evaluation yields:

- $y < -2$: $F_Y(y) = 0$
- $-2 \leq y < -1$: $\frac{y+2}{4}$
- $-1 \leq y < 0$: $\frac{1}{4}$
- At $y = 0$: jump of $\frac{1}{2}$
- $0 < y < 1$: $\frac{3}{4}$
- $1 \leq y \leq 2$: $\frac{y+2}{4}$
- $y > 2$: 1

PDF

$$f_Y(y) = \begin{cases} \frac{1}{4}, & -2 < y < -1 \\ \frac{1}{4}, & 1 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

With:

$$P(Y=0)=\frac{1}{2}$$

Q7. Stock Model Approximation

Setup

$$S_{1000}=S_0u^kd^{1000-k}$$

Condition:

$$S_{1000}\geq 1.30S_0$$

Log Transformation

$$k\ln u+(1000-k)\ln d\geq \ln(1.30)$$

Solve for k

$$k\geq \frac{\ln(1.30)-1000\ln d}{\ln u-\ln d}\approx 469.4\Rightarrow k\geq 470$$

Distribution

$$k\sim \text{Binomial}(1000,0.52)$$

Final Probability

$$P(k\geq 470)=\sum_{k=470}^{1000}\binom{1000}{k}(0.52)^k(0.48)^{1000-k}$$

Q8. Exponential Distribution

Given

$$\lambda = \frac{1}{2}, \quad f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10|X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

CDF

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

Integration Region

$$-z \leq x \leq z, \quad -\sqrt{z^2 - x^2} \leq y \leq \sqrt{z^2 - x^2}$$

CDF Expression

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y)\, dy\, dx$$

PDF via Differentiation

Using Leibniz rule:

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x)\left[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})\right]}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of Maxima

Definitions

$$M_n = \max(X_1, \ldots, X_n), \quad M_{n+1} = \max(M_n, X_{n+1})$$

CDF of M_n

$$P(M_n \le x) = F(x)^n$$

Joint CDF

Case 1: $x \le y$

$$P(M_n \le x, M_{n+1} \le y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \le x, M_{n+1} \le y) = F(y)^{n+1}$$

Final Result

$$P(M_n \le x, M_{n+1} \le y) = \begin{cases} F(x)^n F(y), & x \le y \\ F(y)^{n+1}, & x > y \end{cases}$$

End of Scribe

If you want, I can now:

- convert this into **clean LaTeX (exact exam-ready format)**, or
- compress it into a **1-page revision cheat sheet** (super useful before exams).